

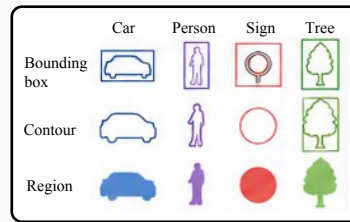
Natural Images



Distribution Gap



Semantic categories



Object geometry

Task Gap

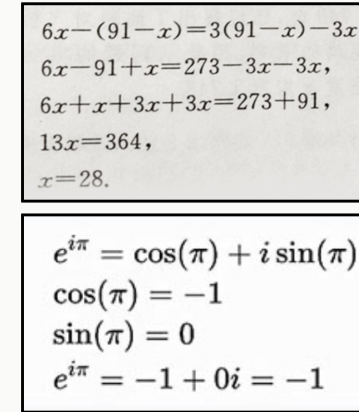
Document Images



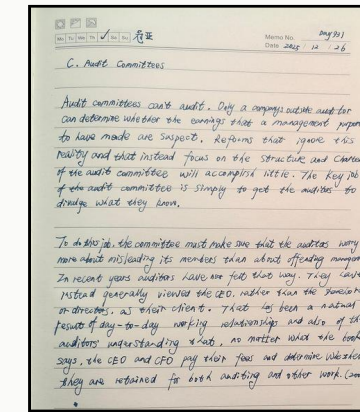
Scientific paper



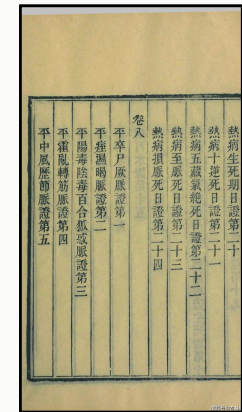
Invoice



Formula



Handwriting



Ancient text

Document-Native Requirements Unmet by Existing Vision Foundation Models



Character strokes



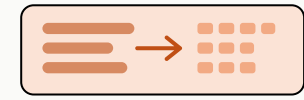
Glyph details



Layout



Reading order



Fine-grained text

Representation Mismatch



Existing Vision Foundation Models

ImageNet



Object semantic

CLIP



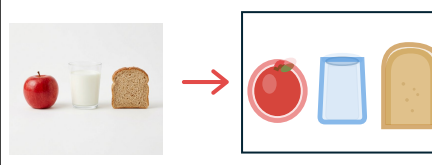
Global alignment

DINO



Semantic feature

SAM



Region boundary

Document-native Representation



Visual-Text Foundation Model

Text Generation



MonkeyOCRv2

Pixel Reconstruction



Document-native visual representation

Text
Recognition

Formula
Recognition

Text
Detection

Document
Parsing

Document
VQA